

DiagHer: When Three Engineering Students Decided to Hack Healthcare

A Journey from Classroom to Changing Lives

The Spark That Started It All

Picture this: Three fourth-year computer engineering students at Khwopa College of Engineering, buried in textbooks and code, when a simple conversation changed everything.

"Have you ever noticed how doctors sometimes don't take women's pain seriously?" Isha asked during one of our lunch breaks.

Jeeni nodded, sharing her own experience of being dismissed when she complained about severe menstrual pain. "They told me it was 'normal' and to just take painkillers."

Jenisha chimed in with statistics she'd been reading: "Did you know that women are 50% more likely to be misdiagnosed during a heart attack? And 80% of medication side effects occur in women, yet most drugs are tested primarily on men."

That night, as we sat surrounded by circuit diagrams and programming assignments, we realized something profound: **We weren't just studying to build systems but we were learning to build solutions that could save lives.**

When SHEQUAL Foundation Called

When the SHEQUAL Foundation announced Nepal's Largest All-Female Hackathon with the theme "Hacking for Empowered Women," we knew this was our moment. The universe was literally calling us to action.

The foundation's mission resonated deeply with us ; revolutionizing the ecosystem by ensuring equal perspectives for both men and women. As young women in engineering, we'd experienced firsthand how different perspectives can lead to breakthrough innovations.

We chose the "Health and Lifestyle" subtheme because we believed that healthcare should be the great equalizer, not another space where gender bias thrives.

Meet Team CodeRed

Why CodeRed? Because in medical emergencies, every second counts. But we also realized that women's health has been in a state of "code red" for decades and nobody was paying attention.

Jeeni Shrestha- The Visionary

"I've always believed that technology should serve humanity, not the other way around." Jeeni brings the perfect blend of technical expertise and empathy. Her experience with machine learning and her passion for social impact made her the driving force behind DiagHer's AI architecture.

Jenisha Shrestha - The Innovator

"If we can teach machines to recognize patterns in data, why can't we teach them to recognize patterns in pain?" Jenisha's background in data science and her knack for seeing solutions where others see problems made her our go-to person for developing DiagHer's gender-aware diagnostic algorithms.

Isha Bhattarai- The Bridge-Builder

"The best technology is invisible and it just makes life better." With her expertise in full-stack development and user experience design, Isha ensures that DiagHer isn't just powerful but also accessible and intuitive for healthcare providers. Her talent lies in translating complex AI diagnostics into seamless, human-centered experiences that feel natural in high-pressure medical environments.

The Birth of DiagHer

The Problem That Kept Us Up at Night

We dove deep into research and discovered some shocking truths:

- **70% of women** report feeling dismissed by healthcare providers
- It takes **5 times longer** for women to receive autoimmune disease diagnoses
- **Heart disease symptoms** in women are often misattributed to anxiety or stress **Medication**
- **Dosages** are primarily based on male physiology

We realized that the healthcare system wasn't broken rather it was **designed without women in mind**.

The Solution That Sparked Hope

DiagHer isn't just another AI tool. It's a revolution disguised as software.

Here's how our system works:

1. **Gender-Aware Diagnosis:** When a doctor enters symptoms, DiagHer provides gender-specific disease likelihoods based on the latest research and real-world data.
2. **Medication Safety Net:** Before prescribing, DiagHer analyzes medications for gender-specific side effects and suggests safer alternatives when available.
3. **Continuous Learning:** Every diagnosis and treatment outcome feeds back into the system, making it smarter and more accurate over time.

But here's the magic: **DiagHer learns from every interaction, becoming more accurate and equitable with each use.**

Day 1: The Foundation Rush

Coffee in hand, we hit the ground running. The energy in the room was electric with forty-eight brilliant women ready to change the world in just 48 hours. We started by building the core architecture: Jeeni worked on the AI models, Jenisha tackled the database design, and Isha crafted the user interface.

By hour 6, we had our first breakthrough. Our gender-aware diagnostic algorithm was taking shape, and we could already see the potential impact.

Day 2: The Deep Dive

This is where the real magic happened. At 3 AM, when most teams were struggling with caffeine crashes, we had our "eureka" moment. We successfully integrated gender-specific medical data with our diagnostic algorithm. The first test run gave us chills as it correctly identified that chest pain in a 45-year-old woman had a 78% probability of being heart disease, not anxiety.

We've been living on energy drinks, determination, and the occasional stress-induced laugh. Our fingers are flying across keyboards as we fine-tune every aspect of DiagHer.

Tomorrow: The Final Push

As we write this, it's the night before our final presentations. We're exhausted, exhilarated, and honestly a little nervous. But we're also proud. DiagHer is more than we ever imagined when we started this journey 36 hours ago.

Our demo is ready. Our presentation is polished. Our hearts are full of hope.

Tomorrow, we don't just present a project rather we unveil a future where healthcare understands gender.

What We've Learned So Far

Even before the final results, this hackathon has taught us invaluable lessons:

1. **Innovation happens when diverse perspectives collide** - Our different backgrounds and experiences made DiagHer more robust and comprehensive.
2. **Technology is only as good as the problem it solves** - We weren't building for the sake of building; we were solving a real, urgent problem.
3. **Small teams can create big change** - Three students with laptops and determination can challenge an entire healthcare system.
4. **Pressure creates diamonds** - The 48-hour timeline pushed us to think creatively and execute efficiently.

The Ripple Effect (Already Starting)

Even during the hackathon, we've been amazed by the response:

- **Fellow participants** stopping by our workstation to learn about DiagHer
- **Mentors** offering guidance and expressing genuine excitement about our solution
- **Different background professionals** among the mentors already asking about beta-testing opportunities

Tomorrow's Promise

As we prepare for tomorrow's final presentations, we're not just thinking about winning. We're thinking about the impact DiagHer could have:

Our Vision for the Future

1. **Clinical Trials:** We want to work with private hospitals to test DiagHer in real-world settings.
2. **Expanded Database:** Collaborating with medical researchers to include more gender-specific health data.
3. **Global Scale:** Adapting DiagHer for different healthcare systems and cultural contexts.
4. **AI Ethics:** Ensuring our algorithms are bias-free and continuously improving.

A Message to Our Future Selves

Dear Future Us,

It's 11 AM, and we're about to get a few hours of sleep before the biggest presentation of our academic careers. Whatever happens tomorrow, remember this feeling; the excitement of staying up all night coding for a cause you believed in, the camaraderie of working with your best friends toward a common goal, and the pure joy of seeing your idea come to life.

Remember that you started this journey not because you were the smartest in the room, but because you cared the most about the problem.

Tomorrow, when you stand in front of those judges, remember every woman who has been dismissed, misdiagnosed, or under-treated. You're not just presenting code but you're presenting hope.

Whatever the outcome, you've already won. You've proven that three women with laptops can build something that matters.

With love, determination, and a healthy dose of caffeine, Present Us

The SHEQUAL Foundation Impact

The SHEQUAL Foundation didn't just organize a hackathon but they created a movement. They showed us that when women come together to solve problems, magic happens.

This isn't just about winning a competition. It's about:

- **Empowering** women to take up space in tech
- **Encouraging** innovative solutions to real-world problems
- **Building** a community of change-makers
- **Proving** that diversity drives innovation

Join Us Tomorrow (And Beyond)

DiagHer is more than a project; it's a statement. It says that women's health matters, that gender bias has no place in healthcare, and that technology should serve everyone equally.

Tomorrow, we present. But our journey doesn't end there.

Whether we win or not, we're committed to taking DiagHer forward. We believe in this solution, and we believe in the change it can bring.

Want to be part of this revolution?

- **Healthcare professionals:** Help us test and improve DiagHer
- **Fellow developers:** Contribute to our open-source components
- **Researchers:** Share your data and insights
- **Advocates:** Spread the word about gender-aware healthcare

The Beginning, Not the End

As we write this blog post, preparing for tomorrow's final presentations, we're already thinking beyond the hackathon. DiagHer is our baby, and we're not stopping here.

But the most important thing we've learned is this: **When you combine passion with purpose, technical skills with empathy, and determination with opportunity, you don't just build software but you build hope.**

To every woman reading this: Your pain is valid. Your symptoms matter. Your voice deserves to be heard. And thanks to initiatives like the SHEQUAL Foundation hackathon, the future of healthcare is being written by women who understand your struggles.

To every aspiring engineer: Don't just learn to code but learn to care. Don't just build systems but build solutions. Don't just follow trends but create change.

To our fellow hackers: Keep hacking. Keep building. Keep believing that technology can make the world more equitable, one line of code at a time.

Tomorrow, we present DiagHer. But today, we've already changed ourselves.

DiagHer: Because healthcare should understand everyone, not just half the population.

Special thanks to the SHEQUAL Foundation for creating a platform where women's voices are not just heard, but celebrated. Here's to more hackathons, more innovations, and more women changing the world.